MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, INTERFACE, NTDS (ANEW), KEY CODE QDR

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic interface driver module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Driver, interface, NTDS(ANEW)
Family	Digital
Key code	QDR
Size	1A
Weight	35 grams maximum
Failure rate	0.115 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	748 mW (outputs not loaded) at 5.5 Vdc
Input voltage	
Low level	0.0 V min, 0.8 V max
High level	2.0 V min, V_{CC} max
Input current (enable inputs)	
Low level (V_{IL} = 0.0 V)	-5.19 mA max
(V_{IL} = 0.4 V)	-4.8 mA max
High level (V_{IH} = 2.4 V)	80 µA max
(V_{IH} = 5.5 V)	2.0 mA max
Input current (A inputs)	
Low level (V_{IL} = 0.0 V)	3.46 mA max
(V_{IL} = 0.4 V)	3.2 mA max
High level (V_{IH} = 2.4 V)	80 µA max
(V_{IH} = 5.5 V)	2 mA max

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output voltage			
Low level	(I_{OL} = 40 mA)	0.45 V max	
High level	(I_{OH} = -27 mA)	2.7 V min	
Open circuit voltage			
V_{CC} = 5.25 V		4.5 V max	
Output drive current			
Low level	(V_{OL} = 0.45 V)	40 mA	
High level	(V_{OH} = 2.7 V)	-27 mA	
Output impedance (Enable $\leq$ 0.8 V)		100 kΩ min	
Propagation delay time			
t_{PLH} (data to output), C_L = 150 pF		5 ns min,	54 ns max
t_{PHL} (data to output), C_L = 150 pF		5 ns min,	52 ns max
t_{PLH} (enable to output)		5 ns min,	45 ns max
t_{PHL} (enable to output)		5 ns min,	45 ns max

1.3 Module function. This module contains four independent interface line drivers that meet the requirements of MIL-STD-1397 for type C (NTDS ANEW) output voltage and current levels. Each amplifier is provided with a separate enable input. The amplifier outputs are tri-state.

1.3.1 Input to output relationship. Each interface driver has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Inputs		Output
EN	A	Y
H	H	H
H	L	L
L	X	Off state (high Z)

NOTES:
1. H = High voltage level
2. L = Low voltage level
3. X = Don't care

1.3.2 Parallel outputs. Paralleling of output circuits for the wired-OR operation is permissible.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

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STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1397(SHIPS) - Input/output Interfaces, Standard Digital Data, Navy Systems.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.115 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QDR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/241-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

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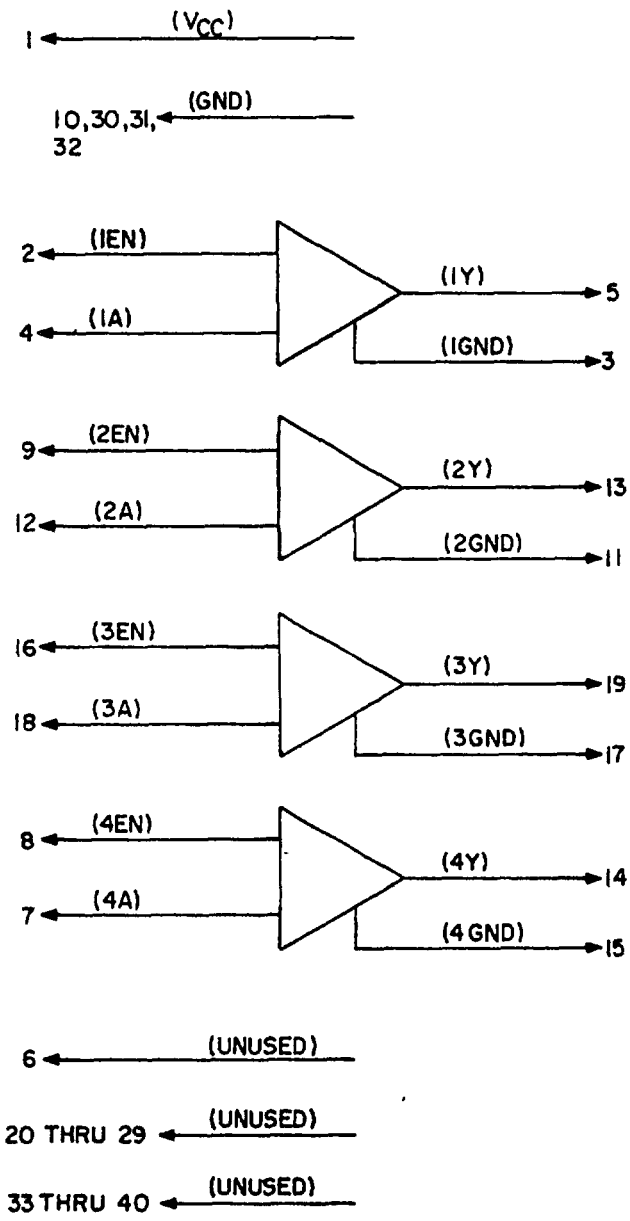


FIGURE 1. QDR module function diagram, driver, interface (NTDS anew).

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Frame isolation	III	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B1-B10	Ground pin continuity	IV	X	0	0.1	0	0.1	0	0.1	0	0.1	Ω
C1-C19	Unused pin isolation	V	X	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
D1-D4	V_{out} (low loaded)	VI	S	0.01	0.90	0.01	0.90	0.01	0.90	0.01	0.90	V
E1-E4	Output short circuit current	VII	X	-50	-250	-50	-250	-50	-250	-50	-250	mA
F1-F8	Input breakdown current	VIII	S	0	2	0	2	0	2	0	2	mA
G1-G4	Static output voltage, low level	IX	X	0.0	0.45	0.0	0.45	0.0	0.45	0.0	0.45	V
H1-H4	Static output voltage, high level	X	X	2.7	4.5	2.7	4.5	2.7	4.5	2.7	4.5	V
J1-J4	Static output voltage, high level, open circuited (V_{CC}=5.25V)	XI	X	2.7	4.5	2.7	4.5	2.7	4.5	2.7	4.5	V
J5-J8	Static output voltage, high level, open circuited (V_{CC}=5.50V)	XI	X	2.7	4.75	2.7	4.75	2.7	4.75	2.7	4.75	V
K1-K4	Output terminal impedance with power removed (+7V on line)	XII	S	0	70	0	70	0	70	0	70	µA
K5-K8	Output terminal impedance with power removed (+3 V on line)	XII	S	0	30	0	30	0	30	0	30	µA
L1-L8	Input current, high level	XIII	X	0	80	0	80	0	80	0	80	µA
M1,M3, M5,M7	Input current, low level enable inputs	XIV	X	-0.1	-5.19	-0.1	-5.19	-0.1	-5.19	-0.1	-5.19	mA
M2,M4 M6,M8	Input current, low level A inputs	XIV	X	-0.1	-3.46	-0.1	-3.46	-0.1	-3.46	-0.1	-3.46	mA

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1	Power supply current, low level	XV	S	40	136	40	136	40	136	40	136	mA
N2	Power supply current, high level	XV	S	15	47	15	47	15	47	15	47	mA
P1-P4	Transition time (t_{TLH}) C_L = 50 pF	XVI	S	5	85	5	85	5	89	5	89	ns
P5-P8	Transition time (t_{TLH}) C_L = 150 pF	XVI	Q	5	90	5	90	5	95	5	95	
Q1-Q4	Transition time (t_{THL}) C_L = 50 pF	XVI	S	5	90	5	90	5	94	5	94	
Q5-Q8	Transition time (t_{THL}) C_L = 150 pF	XVI	Q	5	96	5	96	5	100	5	100	
R1-R4	Propagation delay time (t_{PLH}), data to output, C_L = 50 pF	XVII	S	5	46	5	46	5	50	5	50	
R5-R8	Propagation delay time (t_{PLH}), data to output, C_L = 150 pF	XVII	Q	5	50	5	50	5	54	5	54	
S1-S4	Propagation delay time (t_{PHL}), data to output, C_L = 50 pF	XVII	S	5	44	5	44	5	48	5	48	
S5-S8	Propagation delay time (t_{PHL}), data to output, C_L = 150 pF	XVII	Q	5	48	5	48	5	52	5	52	
T1-T4	Enable recovery time (t_{ERL}), enable to output	XVIII	S	5	43	5	43	5	45	5	45	
U1-U4	Enable recovery time (t_{ERH}), enable to output	XVIII	S	5	43	5	43	5	45	5	45	ns
V1-V4	Output crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
W1-W4	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XXI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
Z1-Z8	Input capacitance	XXII	Q	3.0	15.0	3.0	15.0	3.0	15.0	3.0	15.0	pF

NOTES:
1. (X) Parameters shall be 100% tested on all modules.
2. (S) Parameters shall be sample acceptance tested (1% or two modules from each inspection lot, whichever is greater), at 25°C.
3. (Q) Parameters shall be tested during qualification testing only.

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3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that 5.0 Vdc ±5% is required for test condition J5-J8 which meets the requirements of MIL-STD-1397(SHIPS) for type "C" (NTDS ANEW) interface for guaranteed maximum no load output voltage of 4.5 V.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 Designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
Ohmmeter	Item R01	1
DC voltmeter	Item V04	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitance bridge	Item C03	1
Capacitance meter	Item C05	1
Resistors		
51.1Ω ±1%, 1/4 W		1
127Ω ±1%, 1/4 W		4
20 kΩ ±1%, 1/4 W		1
100Ω ±1%, 1/4 W		4
Value as required for figure 5, note 2		4
51Ω ±5%, 1 W		2
Capacitors		
Value as required for figure 5, note 4		4
Value as required for figure 5, note 5		4
Diode		
1N4150 or equivalent		4

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F where applicable, followed by the remaining tests in the sequence.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E, G through W, and Y shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test conditions Y shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E, G through W, and Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests E, and G through Z shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

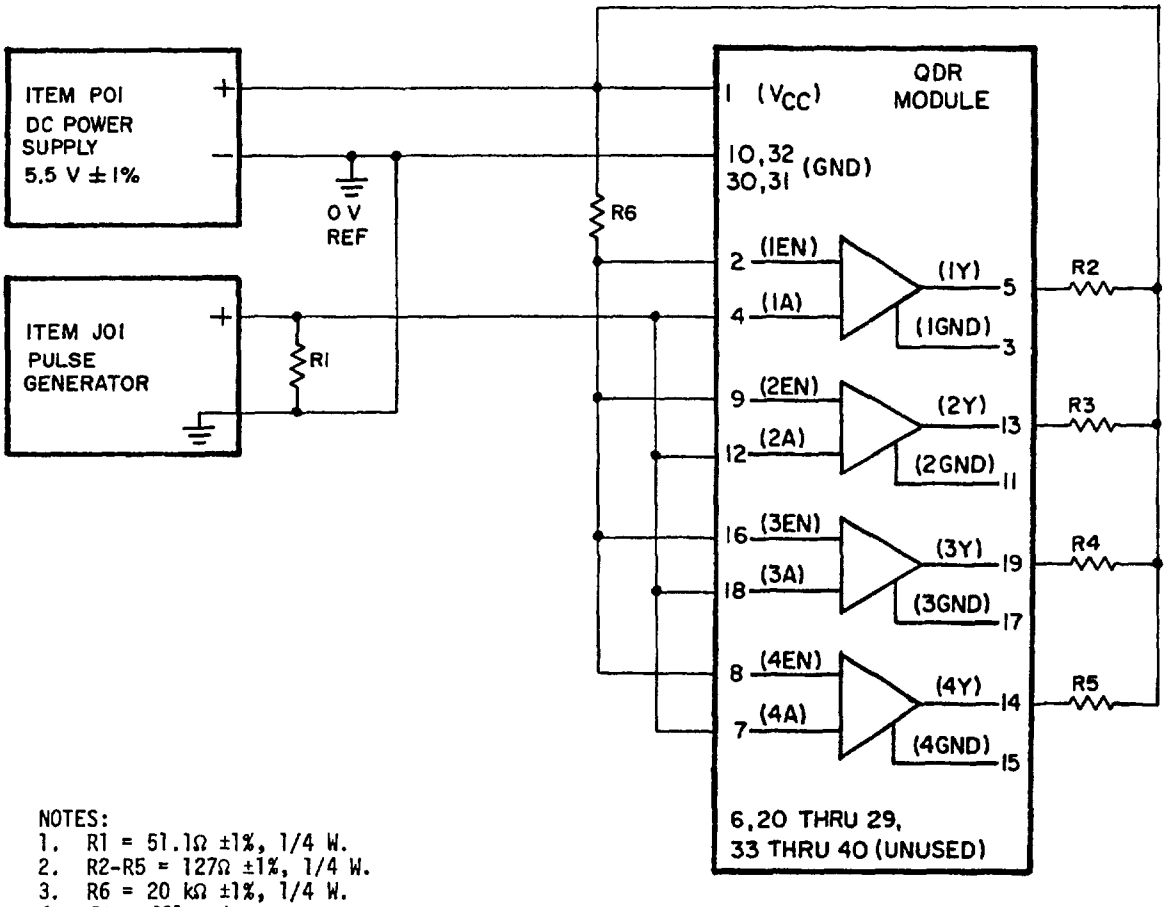
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N033)

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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. R2-R5 = 127Ω ±1%, 1/4 W.
3. R6 = 20 kΩ ±1%, 1/4 W.
4. Item J01 setup:
   Amplitude = 4.5 V.
   PRP = 1,000 ±100 ns.
   PWA = 500 ±50 ns.
5. Verify all outputs have a high level $\geq$ 2.7 V and low level $\leq$ 0.45 V.
6. All voltages are referenced to ground.

FIGURE 2. Operating temperature and life test circuit.

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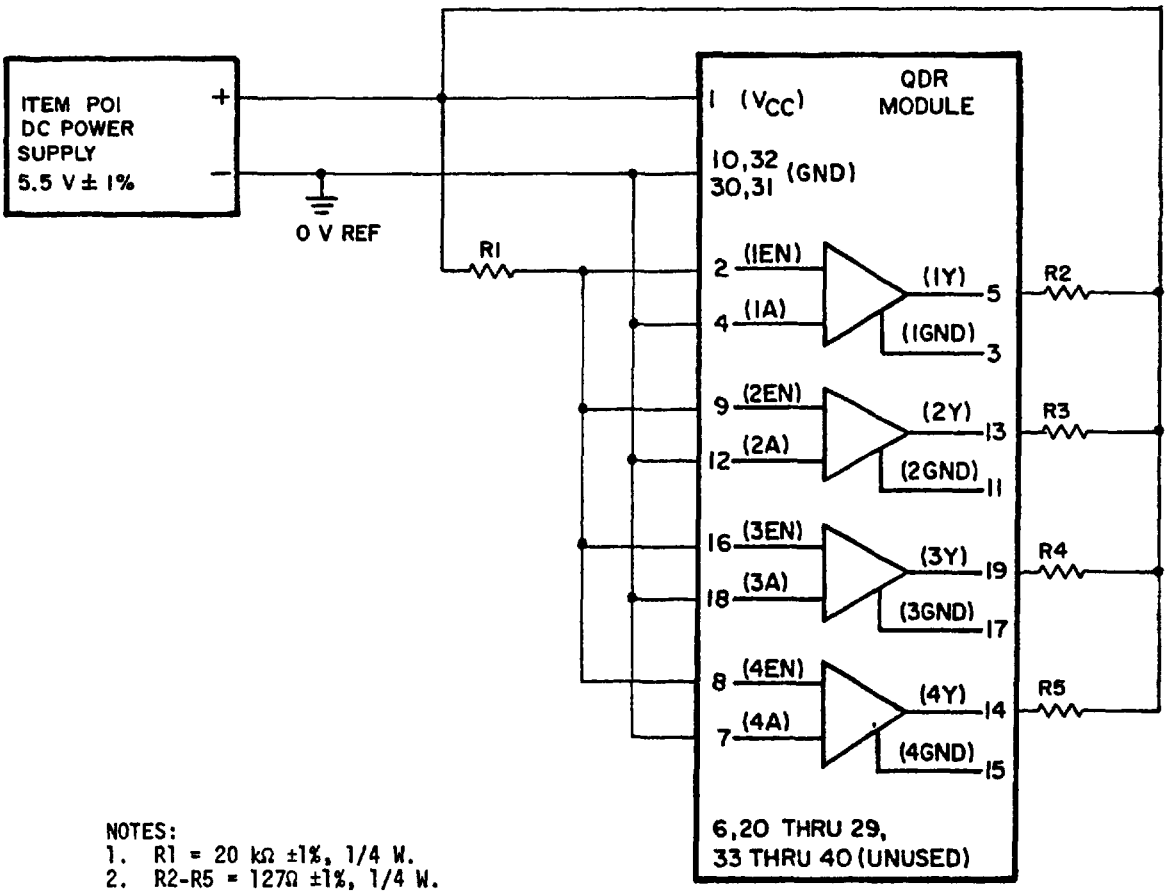


FIGURE 3. Component temperature test circuit.

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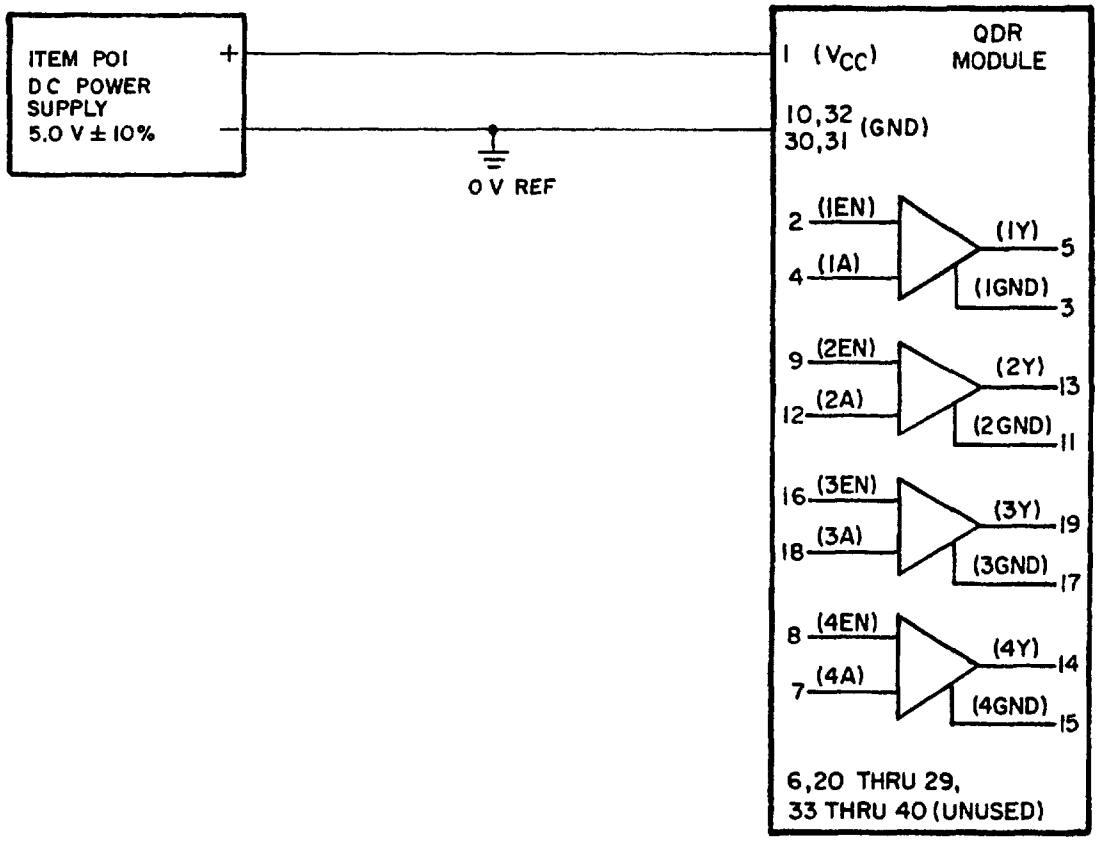
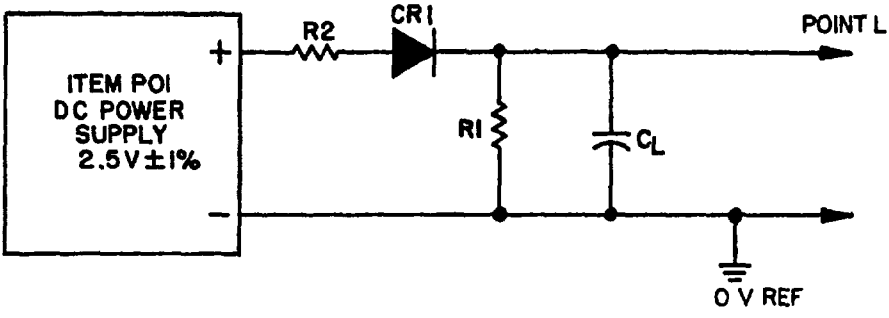


FIGURE 4. Vibration test circuit.

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NOTES:
1. R1 = 100Ω ±1%, 1/4 W.
2. R2 = Select value for a current flow of 40 mA ±5% out of point L with point L held at 0.45 V.
3. CR1 = 1N4150 or equivalent.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P1-P4, Q1-Q4, R1-R4, S1-S4, V1-V4, W1-W4.
5. C_L = 150 pF (including probe and parasitic capacitance) for test conditions P5-P8, Q5-Q8, R5-R8 and S5-S8.
6. C_L is not required when performing dc electrical tests G1-G4 and H1-H4.
7. $C_L \leq 10$ pF for test conditions T1-T4 and U1-U4.

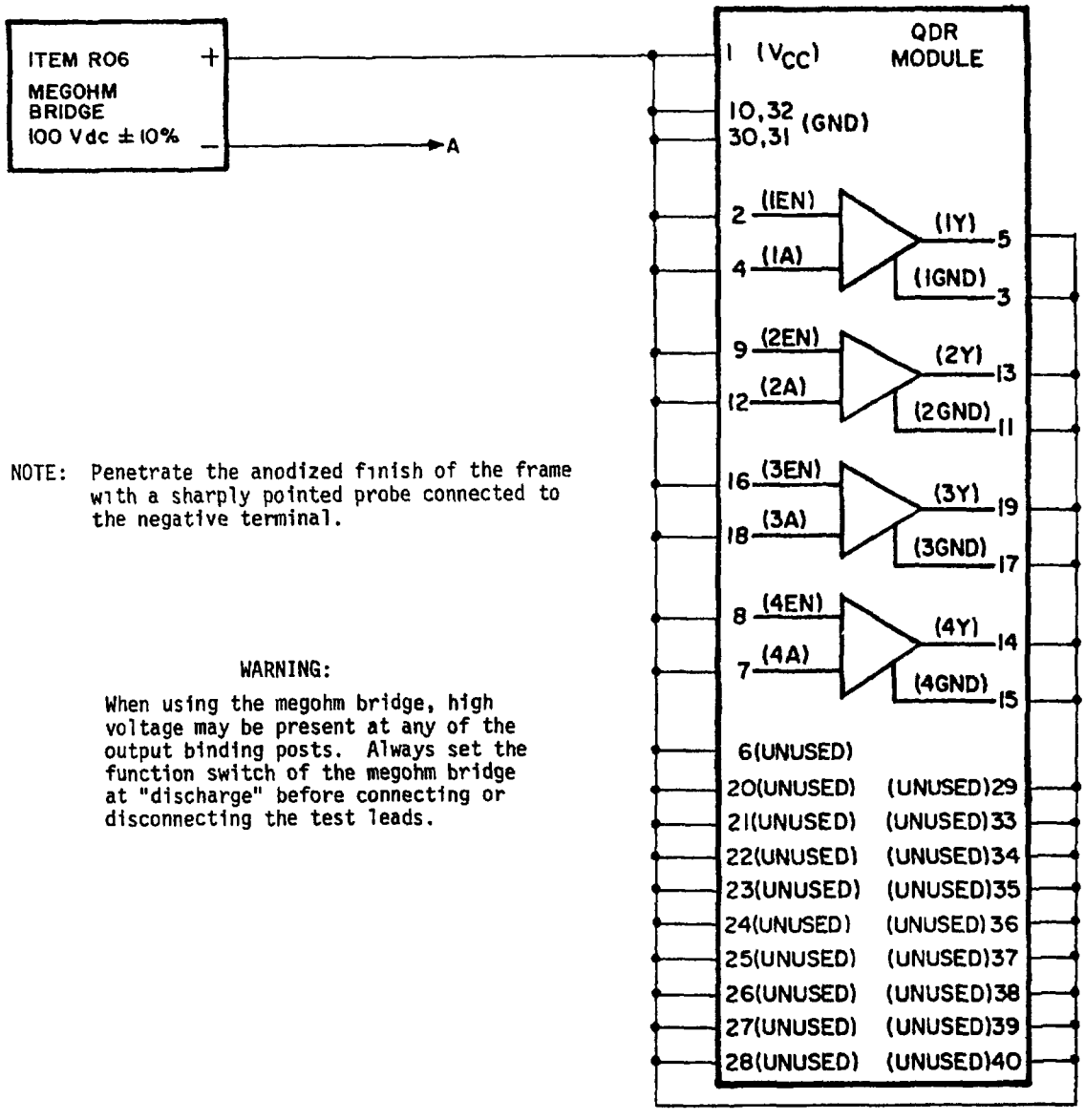
FIGURE 5. QDR logic load circuit.

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TABLE III. Frame isolation test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A	Frame isolation	Frame	10	∞	MΩ	6

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NOTE: Penetrate the anodized finish of the frame with a sharply pointed probe connected to the negative terminal.

WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

FIGURE 6. Frame isolation test circuit for table III.

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TABLE IV. Ground pins continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Ground pins continuity	10	32	0.0	0.1	Ω	7
B2		10	30				
B3		10	31				
B4		32	30				
B5		32	31				
B6	Ground pins continuity	30	31				
B7	Output ground pins continuity	10,32,30,31	3				
B8			11				
B9			17				
B10	Output ground pins continuity	10,32,30,31	15	0.0	0.1	Ω	7

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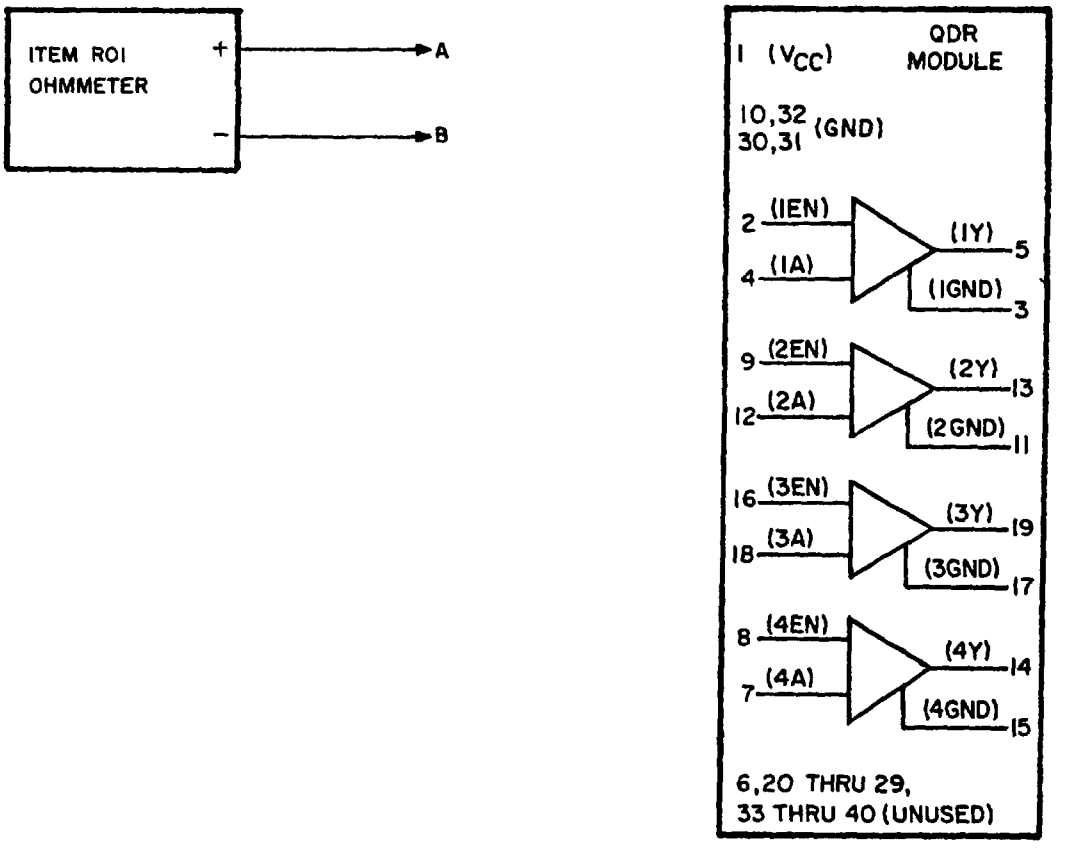


FIGURE 7. Ground pins continuity test circuit for table IV.

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TABLE V. Unused pin isolation test.

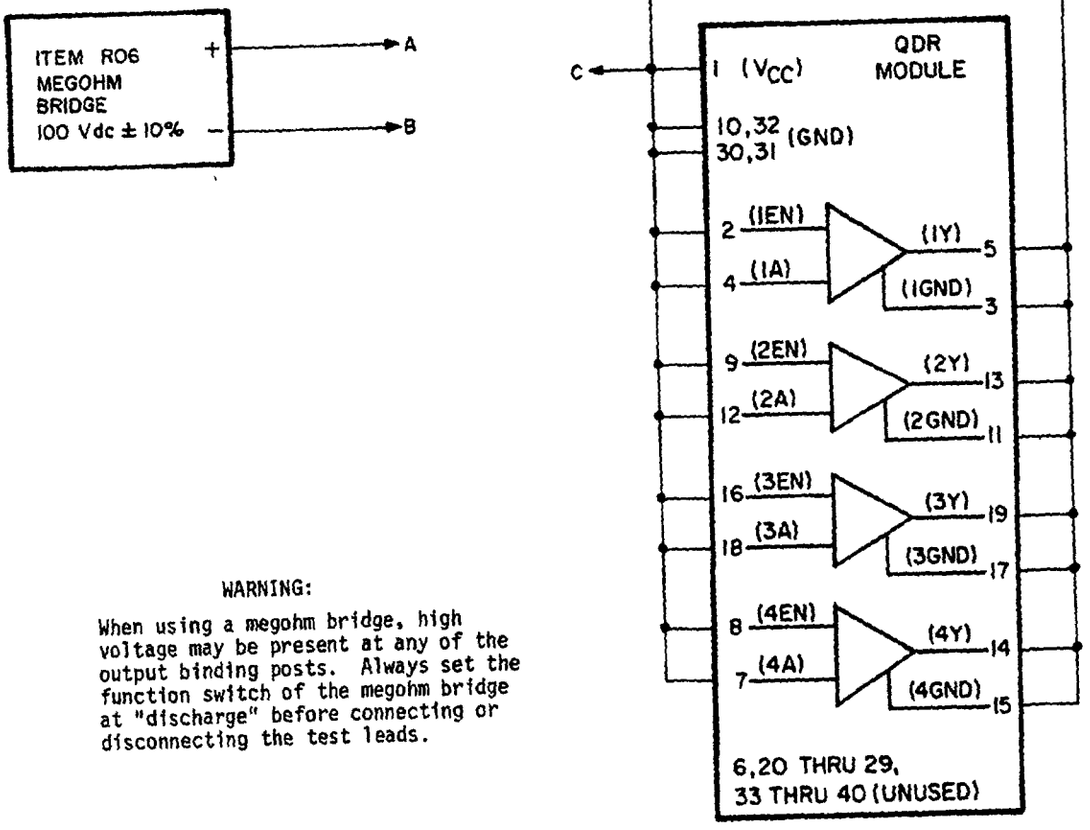
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		B to pin	A to pin	Min	Max		
C1	Unused pin isolation	6,20 thru 29, 33 thru 40	C	10	∞	MΩ	8
C2		20 thru 29, 33 thru 40	6				
C3		21 thru 29, 33 thru 40	20				
C4		22 thru 29, 33 thru 40	21				
C5		23 thru 29, 33 thru 40	22				
C6		24 thru 29, 33 thru 40	23				
C7		25 thru 29, 33 thru 40	24				
C8		26 thru 29, 33 thru 40	25				
C9		27,28,29, 33 thru 40	26				
C10		28, 29, 33 thru 40	27				
C11		29,33 thru 40	28				
C12		33 thru 40	29				
C13	Unused pin isolation	34 thru 40	33	10	∞	MΩ	8

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TABLE V. Unused pin isolation test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		B to pin	A to pin	Min	Max		
C14	Unused pin isolation	35 thru 40	34	10	∞	MΩ	8
C15		36 thru 40	35				
C16		37 thru 40	36				
C17		38,39,40	37				
C18		39,40	38				
C19	Unused pin isolation	40	39	10	∞	MΩ	8

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

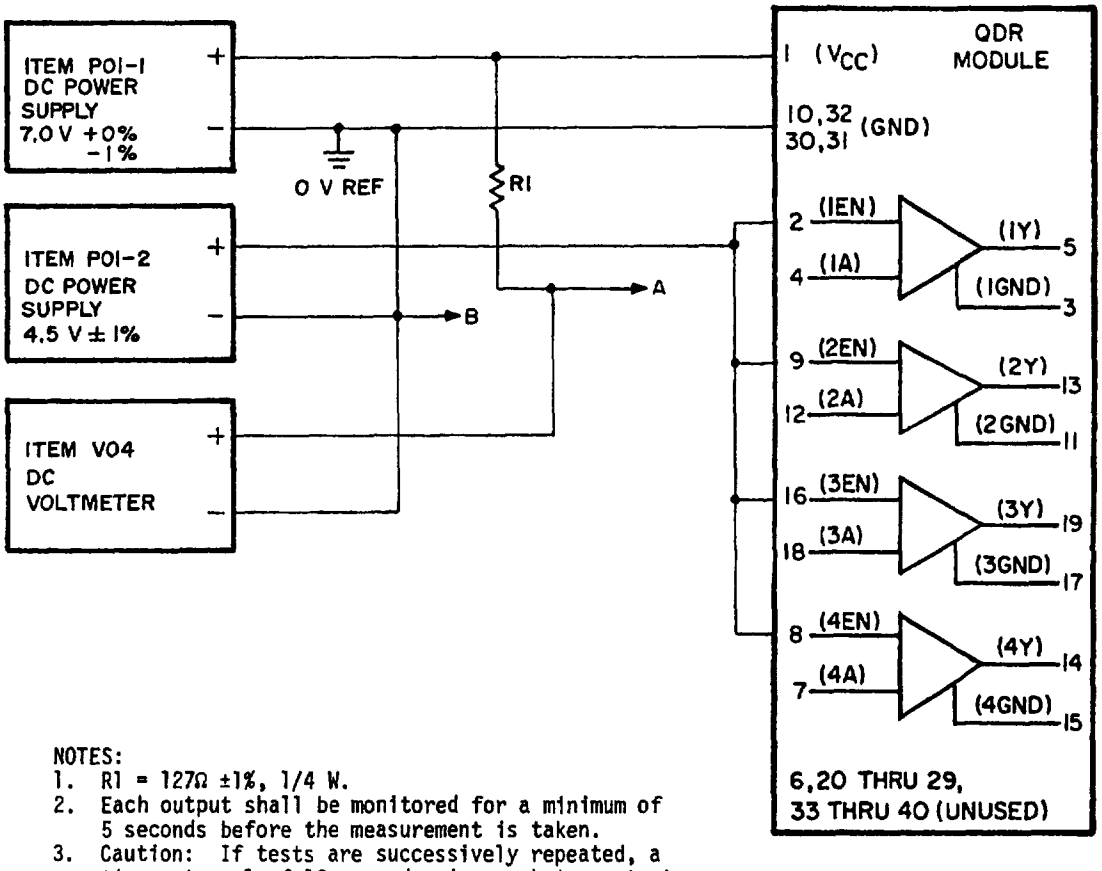
FIGURE 8. Unused pin isolation test circuit for table V.

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TABLE VI. V_{out} (low loaded) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	V_{out} (low loaded)	5	4	0.01	0.90	V	9
D2	↓	13	12	↓	↓	↓	↓
D3		19	18				
D4	V_{out} (low loaded)	14	7	0.01	0.90	V	9

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NOTES:
1. R1 = 127Ω ±1%, 1/4 W.
2. Each output shall be monitored for a minimum of 5 seconds before the measurement is taken.
3. Caution: If tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed.

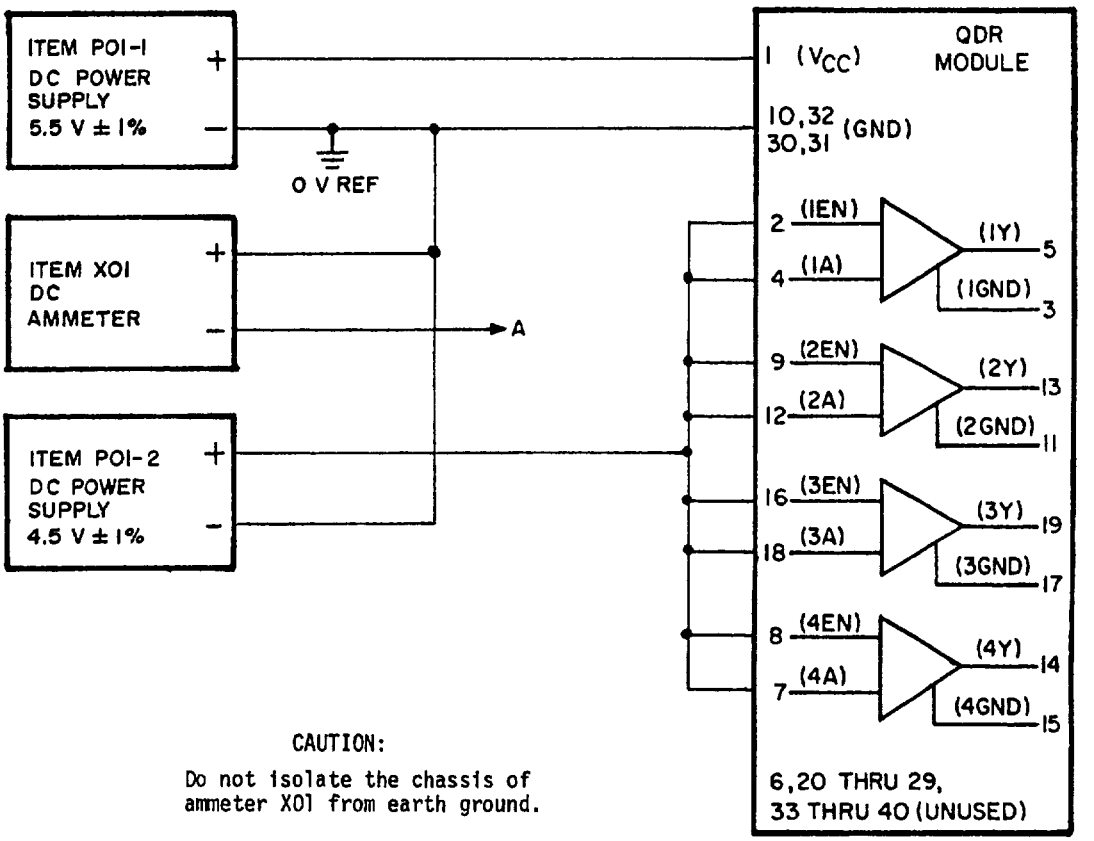
FIGURE 9. V_{OUT} (low loaded) test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Output short circuit current	5	-50	-250	mA	10
E2		13				
E3		19				
E4	Output short circuit current	14	-50	-250	mA	10

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NOTE: When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

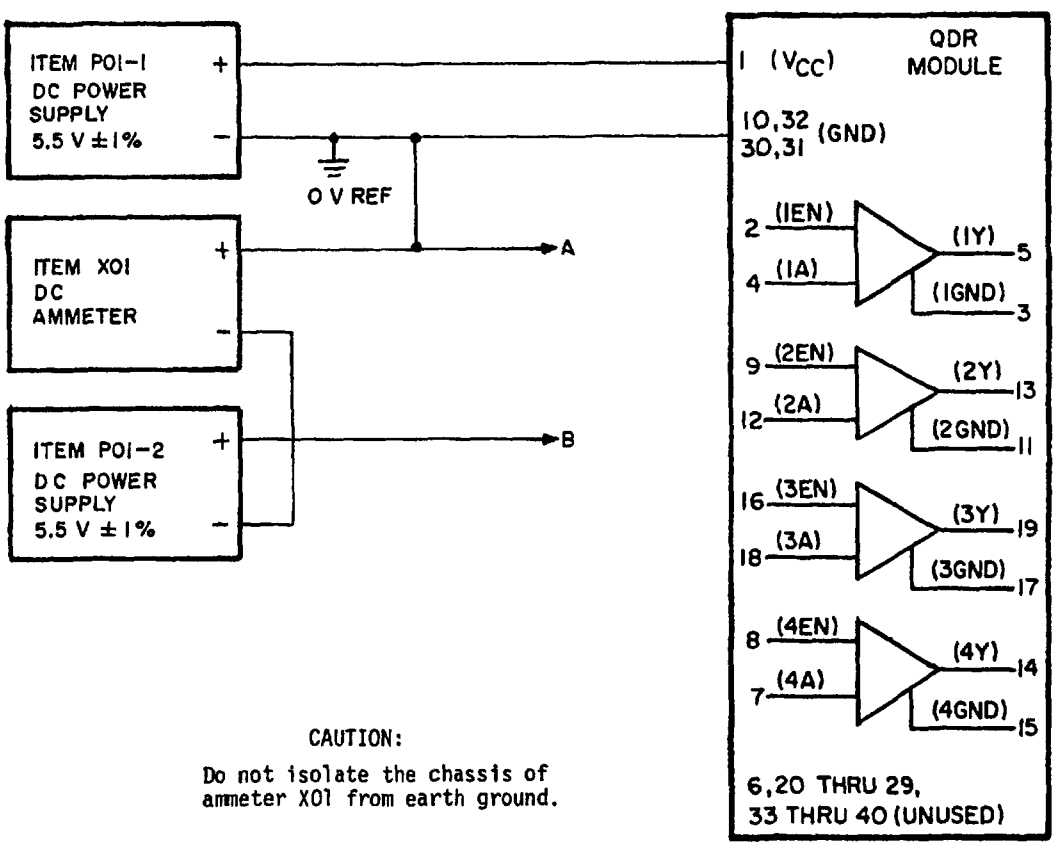
FIGURE 10. Output short circuit current test circuit for table VII.

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TABLE VIII. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	1EN Input breakdown current	4,9,12,16,18, 8,7	2	0.0	2.0	mA	11
F2	1A	2,9,12,16,18, 8,7	4				
F3	2EN	2,4,12,16,18, 8,7	9				
F4	2A	2,4,9,16,18, 8,7	12				
F5	3EN	2,4,9,12,18, 8,7	16				
F6	3A	2,4,9,12,16, 8,7	18				
F7	4EN	2,4,9,12,16, 18,7	8				
F8	4A Input breakdown current	2,4,9,12,16, 18,8	7	0.0	2.0	mA	11

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NOTE: Power supply, item P01-2, shall be current limited to 5.0 mA.

FIGURE 11. Input breakdown current test circuit for table VIII.

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TABLE IX. Static output voltage (low-level) test.

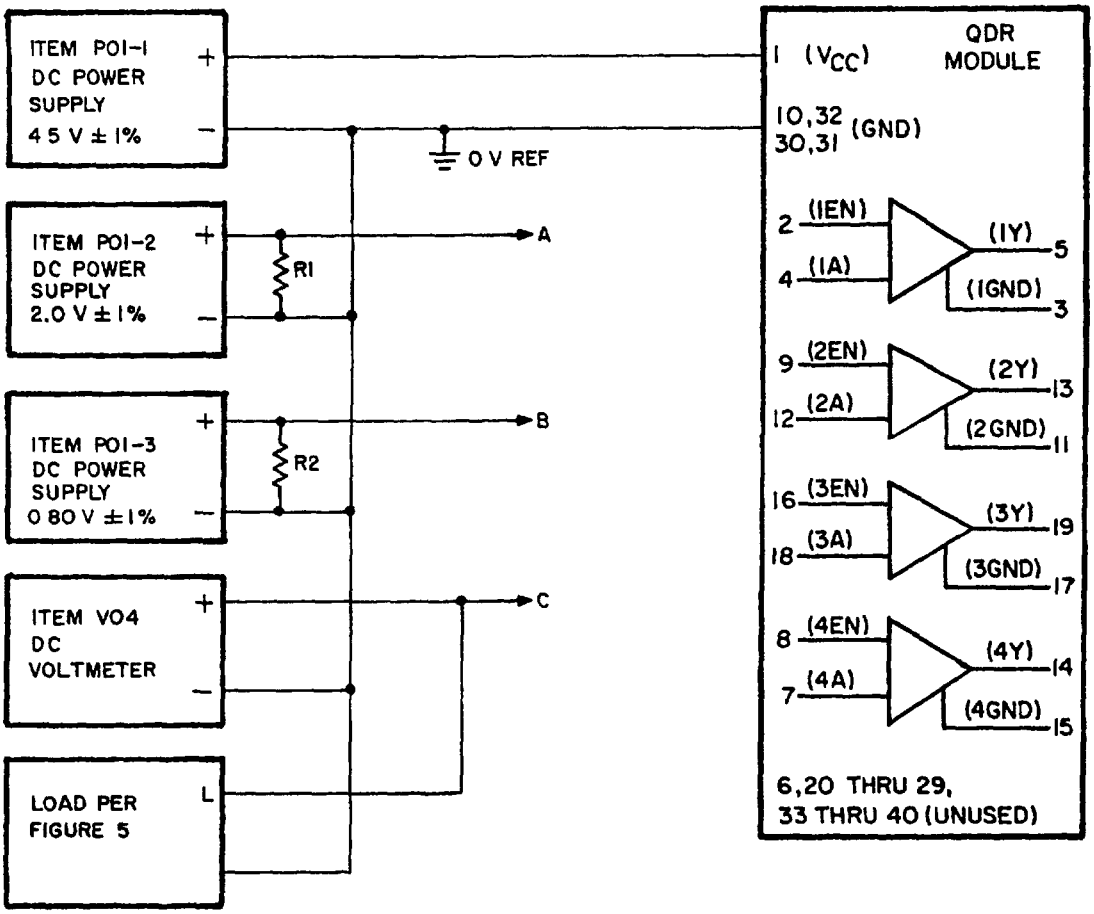
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Static output voltage, low level	2	4	5	0.0	0.45	V	12
G2	↓	9	12	13	↓	↓	↓	↓
G3	↓	16	18	19	↓	↓	↓	↓
G4	Static output voltage, low level	8	7	14	0.0	0.45	V	12

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TABLE X. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Static output voltage, high level	2,4	Open	5	2.7	4.5	V	12
H2	↓	9,12	Open	13	↓	↓	↓	↓
H3	↓	16,18	Open	19	↓	↓	↓	↓
H4	Static output voltage, high level	8,7	Open	14	2.7	4.5	V	12

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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 12. Static output voltage (low-level and high-level) test circuit for table IX and X.

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TABLE XI. Static output voltage (high-level) open circuit test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Static output voltage high level, open circuit, (V_{CC} = 5.25 V)	1	Open	5	2.7	4.5	V	13
J2	↓	1	Open	13	↓	↓	↓	↓
J3	↓	1	Open	19	↓	↓	↓	↓
J4	Static output voltage high level, open circuit, (V_{CC} = 5.25 V)	1	Open	14	↓	4.5	↓	↓
J5	Static output voltage high level, open circuit, (V_{CC} = 5.50 V)	Open	1	5	↓	4.75	↓	↓
J6	↓	Open	1	13	↓	↓	↓	↓
J7	↓	Open	1	19	↓	↓	↓	↓
J8	Static output voltage high level, open circuit, (V_{CC} = 5.50 V)	Open	1	14	2.7	4.75	V	13

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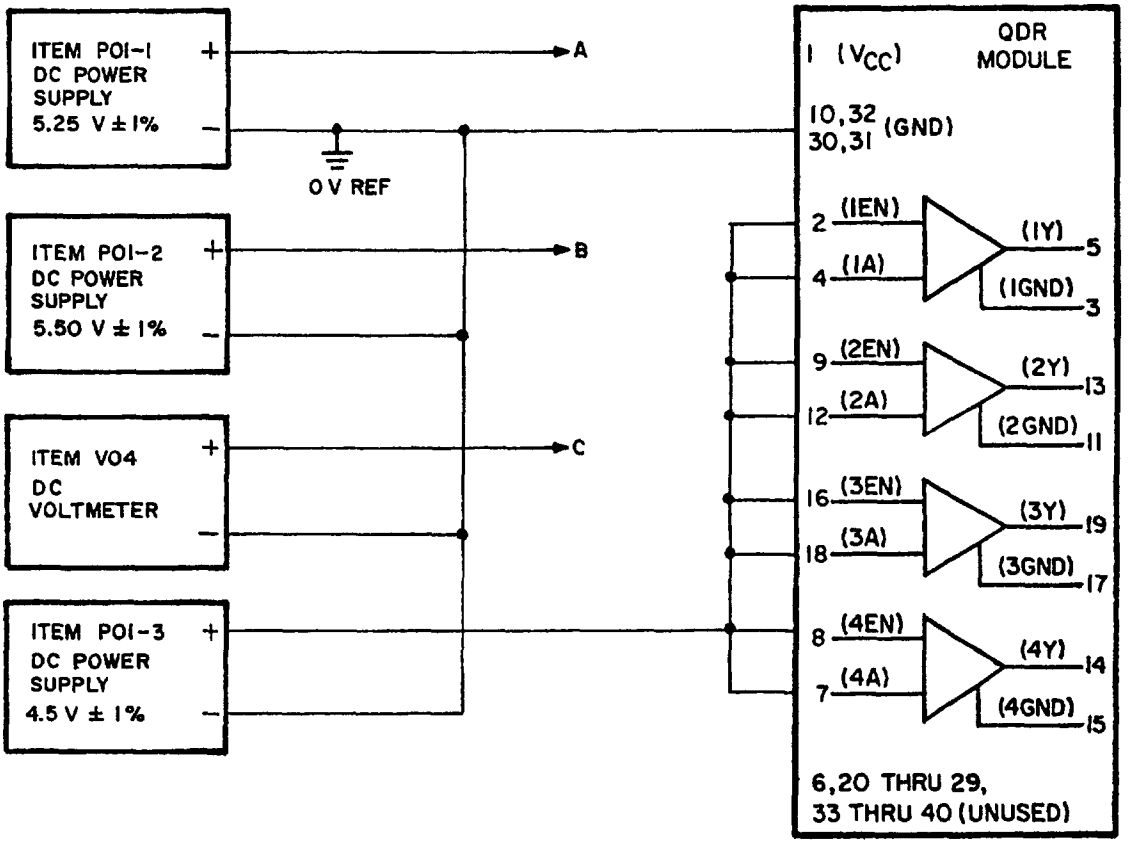


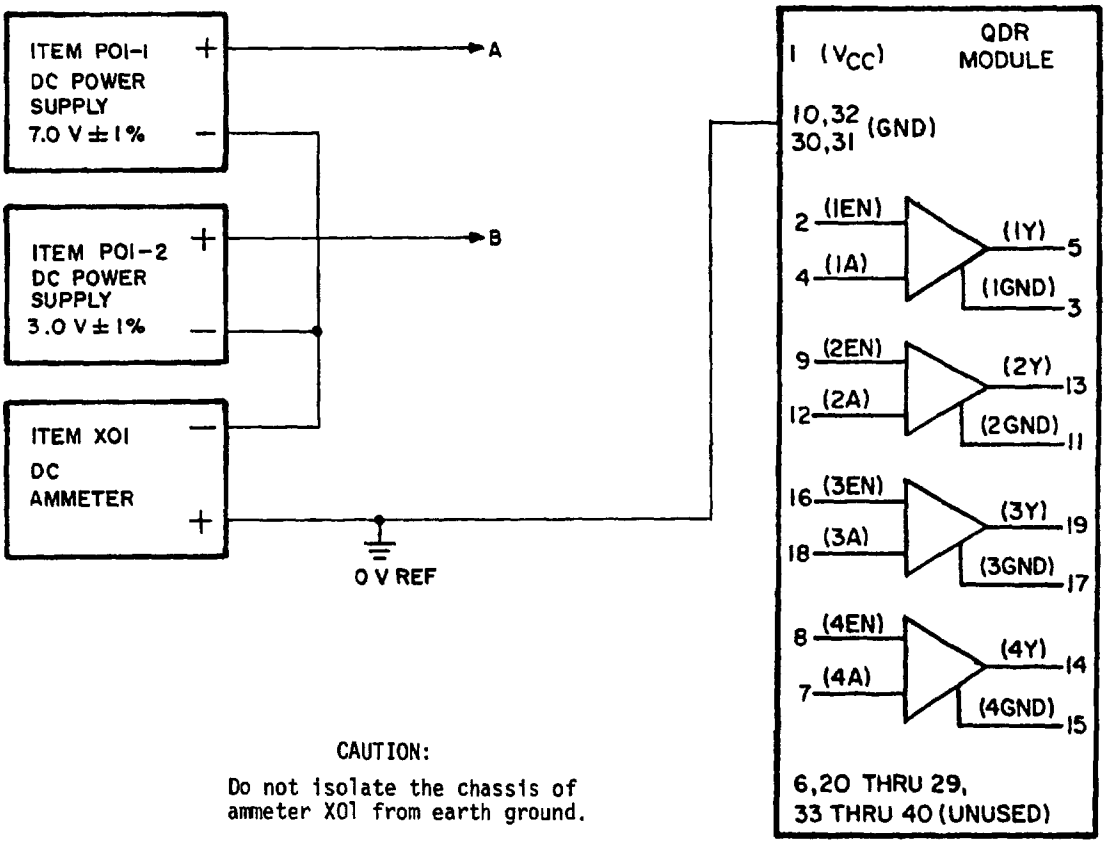
FIGURE 13. Static output voltage (high-level) open circuit test circuit for table XI.

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TABLE XII. Output terminal impedance (to line with module power removed) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Output terminal impedance with power removed, +7 V on line	5	Open	0	70	µA	14
K2		13	Open				
K3		19	Open				
K4	Output terminal impedance with power removed, +7 V on line	14	Open		70		
K5	Output terminal impedance with power removed, +3 V on line	Open	5		30		
K6		Open	13				
K7		Open	19				
K8	Output terminal impedance with power removed, +3 V on line	Open	14	0	30	µA	14

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 14. Output terminal impedance (to line with module power removed) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1EN Input current, high level	4,9,12,16,18, 8,7	2	0	80	µA	15
L2	1A	2,9,12,16,18, 8,7	4				
L3	2EN	2,4,12,16,18, 8,7	9				
L4	2A	2,4,9,16,18, 8,7	12				
L5	3EN	2,4,9,12,18, 8,7	16				
L6	3A	2,4,9,12,16, 8,7	18				
L7	4EN	2,4,9,12,16, 18,7	8				
L8	4A Input current, high level	2,4,9,12,16, 18,8	7	0	80	µA	15

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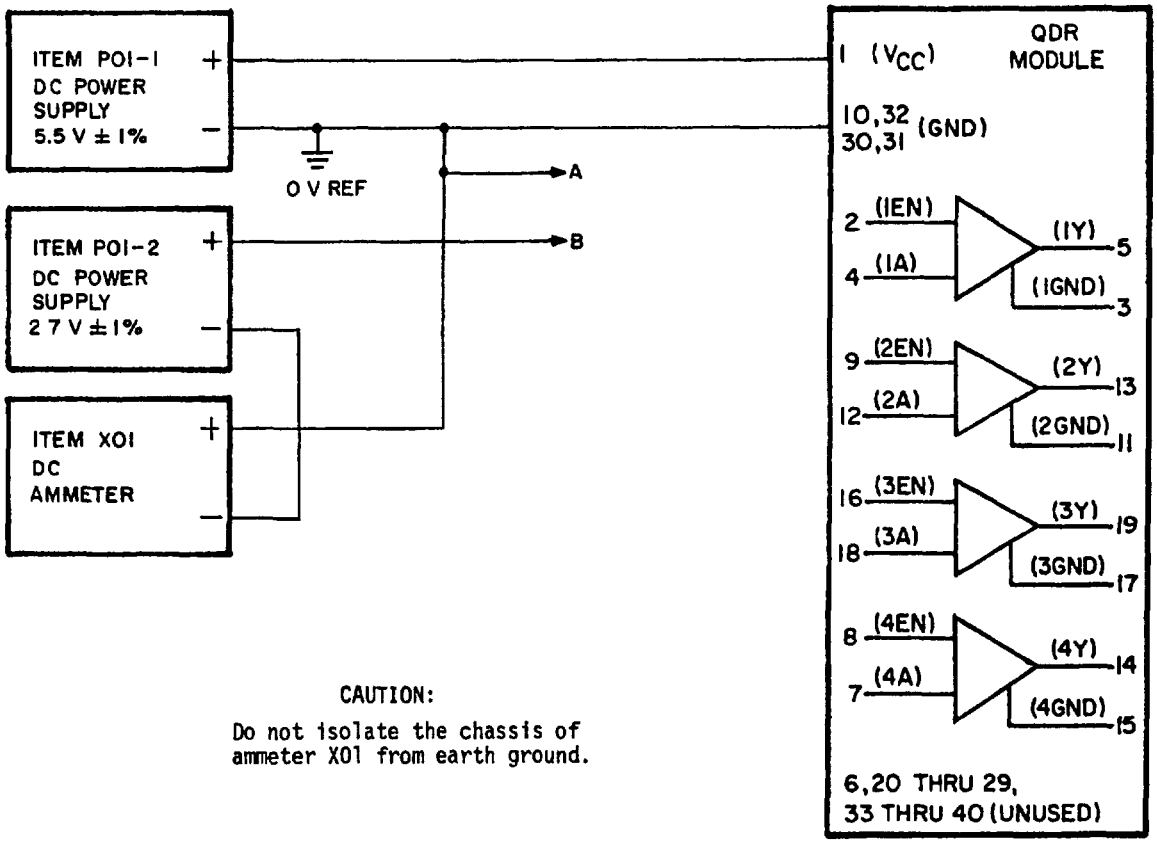


FIGURE 15. Input current (high-level) test circuit for table XIII.

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TABLE XIV. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
M1	1EN Input current, low level	2	-0.1	-5.19	mA	16
M2	1A	4		-3.46		
M3	2EN	9		-5.19		
M4	2A	12		-3.46		
M5	3EN	16		-5.19		
M6	3A	18		-3.46		
M7	4EN	8		-5.19		
M8	4A Input current, low level	7	-0.1	-3.46	mA	16

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Check the source to verify that this is the current version before use.

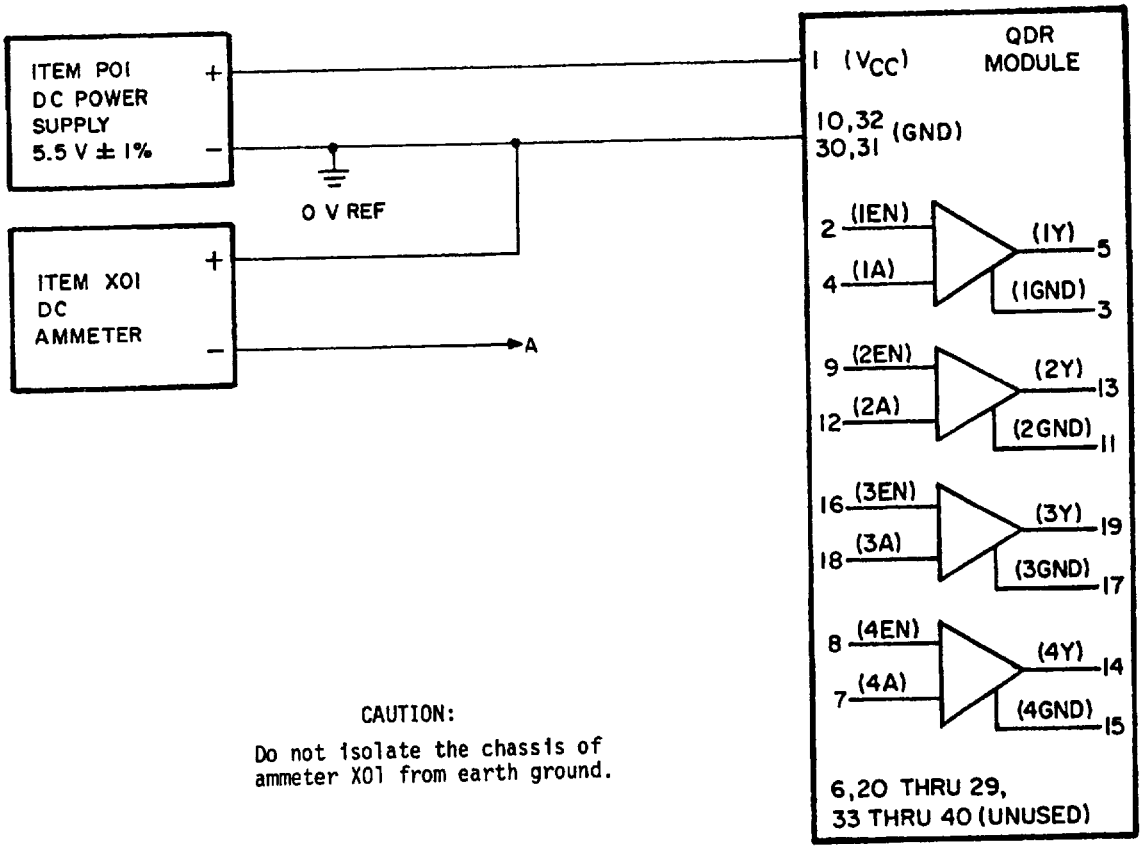


FIGURE 16. Input current (low-level) test circuit for table XIV.

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TABLE XV. Power supply current (low-level and high-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Power supply current, low level output	4,12,18,7	2,9,16,8	40	136	mA	17
N2	Power supply current, high level output	Open	2,4,9,12, 16,18,8,7	15	47	mA	17

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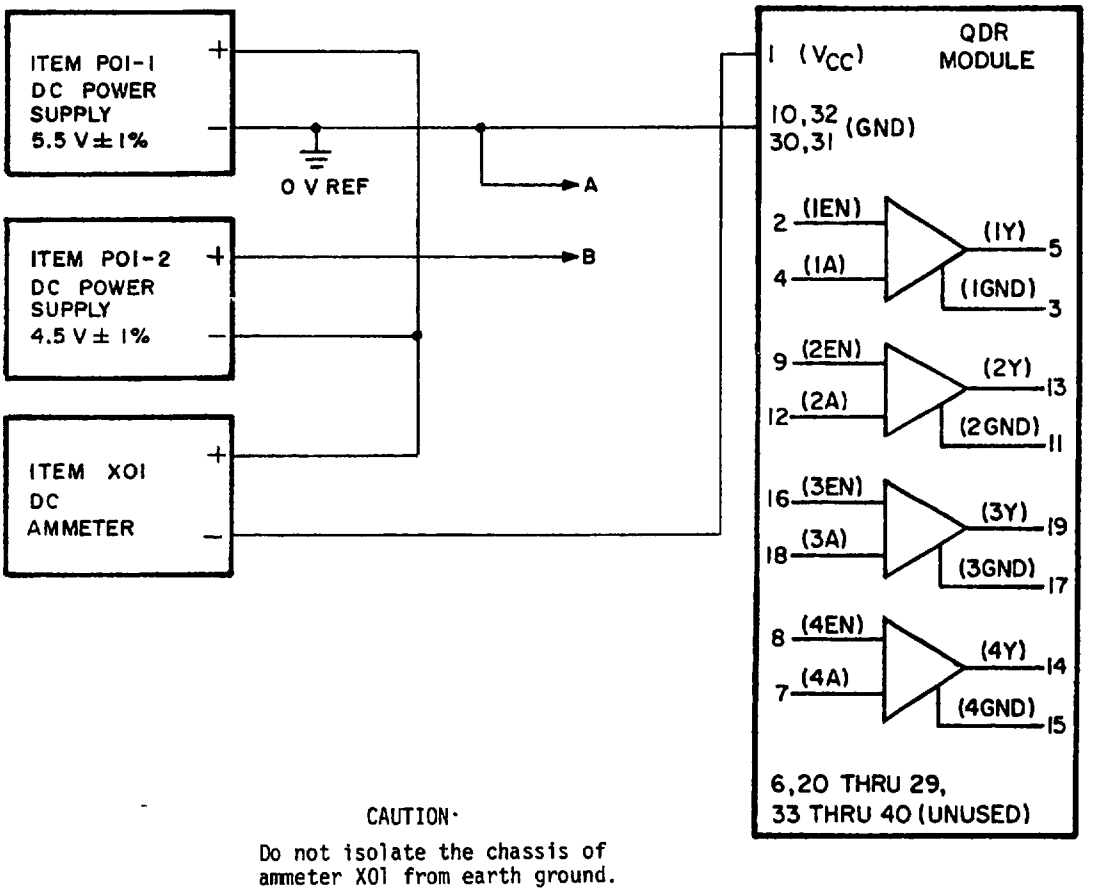


FIGURE 17. Power supply current (low-level and high-level outputs) test circuit for table XV.

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Y Transition time, (t_{TLH}) C_L = 50 pF	2	4	5	Open	5	85	ns	18,19
P2	2Y	9	12	13					
P3	3Y	16	18	19					
P4	4Y Transition time, (t_{TLH}) C_L = 50 pF	8	7	14			85		
P5	1Y Transition time, (t_{TLH}) C_L = 150 pF	2	4	5			90		
P6	2Y	9	12	13					
P7	3Y	16	18	19					
P8	4Y Transition time, (t_{TLH}) C_L = 150 pF	8	7	14			90		
Q1	1Y Transition time, (t_{THL}) C_L = 50 pF	2	4	5			90		
Q2	2Y	9	12	13					
Q3	3Y	16	18	19					
Q4	4Y Transition time, (t_{THL}) C_L = 50 pF	8	7	14			90		
Q5	1Y Transition time, (t_{THL}) C_L = 150 pF	2	4	5			96		
Q6	2Y Transition time, (t_{THL}) C_L = 150 pF	9	12	13	Open	5	96	ns	18,19

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q7	3Y Transition time, (t_{THL}) C_L = 150 pF	16	18	19	Open	5	96	ns	18,19
Q8	4Y Transition time, (t_{THL}) C_L = 150 pF	8	7	14	Open	5	96	ns	18,19

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1A to 1Y Propagation delay time, (t_{PLH}) C_L = 50 pF	2	4	5	Open	5	46	ns	18,20
R2	2A to 2Y	9	12	13					
R3	3A to 3Y	16	18	19					
R4	4A to 4Y Propagation delay time, (t_{PLH}) C_L = 50 pF	8	7	14			46		
R5	1A to 1Y Propagation delay time, (t_{PLH}) C_L = 150 pF	2	4	5			50		
R6	2A to 2Y	9	12	13					
R7	3A to 3Y	16	18	19					
R8	4A to 4Y Propagation delay time, (t_{PLH}) C_L = 150 pF	8	7	14			50		
S1	1A to 1Y Propagation delay time, (t_{PHL}) C_L = 50 pF	2	4	5			44		
S2	2A to 2Y	9	12	13					
S3	3A to 3Y	16	18	19					
S4	4A to 4Y Propagation delay time, (t_{PHL}) C_L = 50 pF	8	7	14			44		
S5	1A to 1Y Propagation delay time, (t_{PHL}) C_L = 150 pF	2	4	5			48		
S6	2A to 2Y Propagation delay time, (t_{PHL}) C_L = 150 pF	9	12	13	Open	5	48	ns	18,20

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test. - Continued.

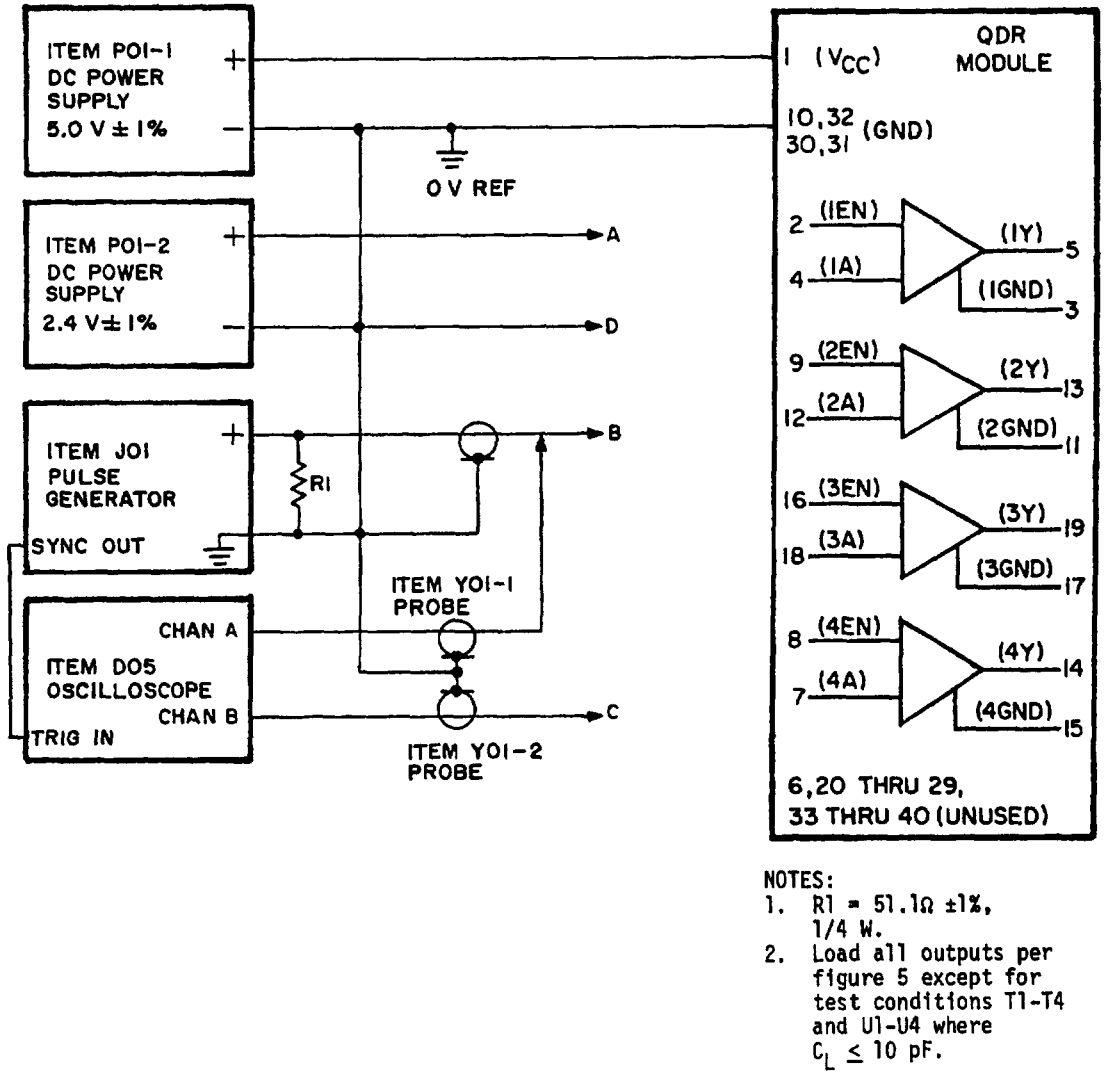
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S7	3A to 3Y Propagation delay time, (t_{PHL}) C_L = 150 pF	16	18	19	Open	5	48	ns	18,20
S8	4A to 4Y Propagation delay time, (t_{PHL}) C_L = 150 pF	8	7	14	Open	5	48	ns	18,20

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TABLE XVIII. Enable recovery time (t_{ERL} and t_{ERH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1EN to 1Y Enable recovery time (t_{ERL})	Open	2	5	4	5	43	ns	18,21
T2	2EN to 2Y	Open	9	13	12				
T3	3EN to 3Y	Open	16	19	18				
T4	4EN to 4Y Enable recovery time (t_{ERL})	Open	8	14	7				
U1	1EN to 1Y Enable recovery time (t_{ERH})	4	2	5	Open				
U2	2EN to 2Y	12	9	13	Open				
U3	3EN to 3Y	18	16	19	Open				
U4	4EN to 4Y Enable recovery time (t_{ERH})	7	8	14	Open	5	43	ns	18,21

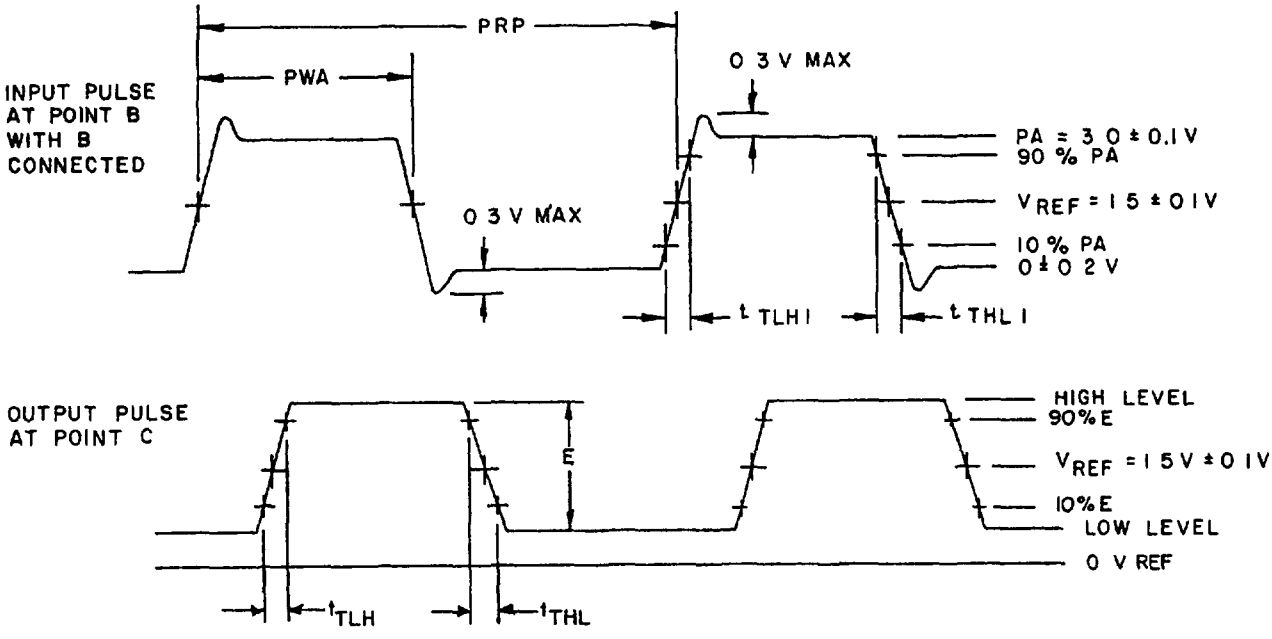
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs per figure 5 except for test conditions T1-T4 and U1-U4 where $C_L \leq 10$ pF.

FIGURE 18. Transition time, propagation delay time and enable recovery time test circuit for tables XVI, XVII and XVIII.

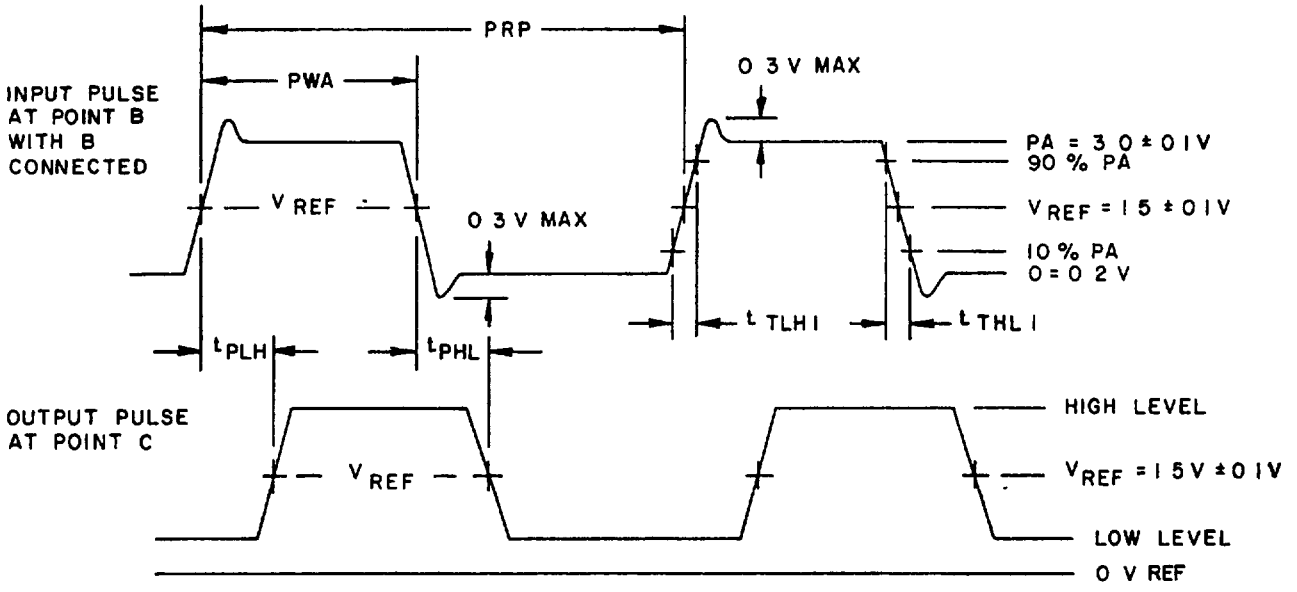
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Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table, verify that the output pulse high level is $\geq$ 2.7 V and the low level is $\leq$ 0.45 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1 and Y01-2 to output pins and between load circuits and output pins shall be less than 2 inches (50.8 mm).
4. All voltages referenced to ground.

FIGURE 19. Transition time test circuit waveforms for table XVI.

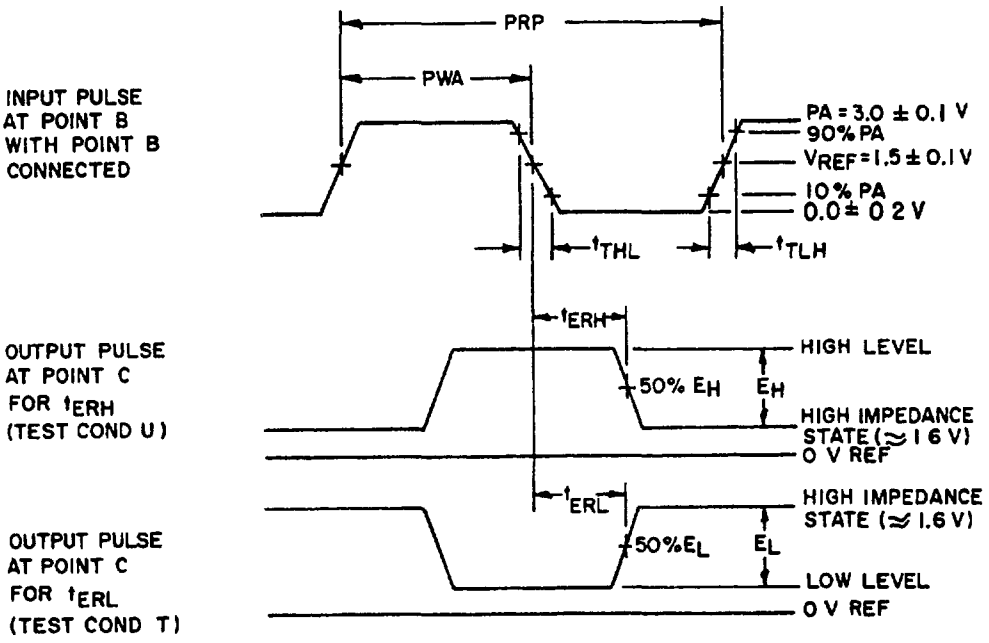
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Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table, verify that the output pulse high level is $\geq$ 2.7 V and the low level is $\leq$ 0.45 V.
3. The lead length of electrical conductors between termination resistor R1 and input pins, probes Y01-1, Y01-2 to output pins and between load circuits and output pins shall be less than 2 inches (50.8 mm).
4. All voltages referenced to ground.

FIGURE 20. Propagation delay time test circuit waveform for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the measurements specified in table XVIII, verify that the output pulse high level is $\geq$ 2.7 V and the low level is $\leq$ 0.45 V.
3. The lead length of electrical conductors between resistor R1 and input pin and between probes YO1-1 and YO1-2 and input and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

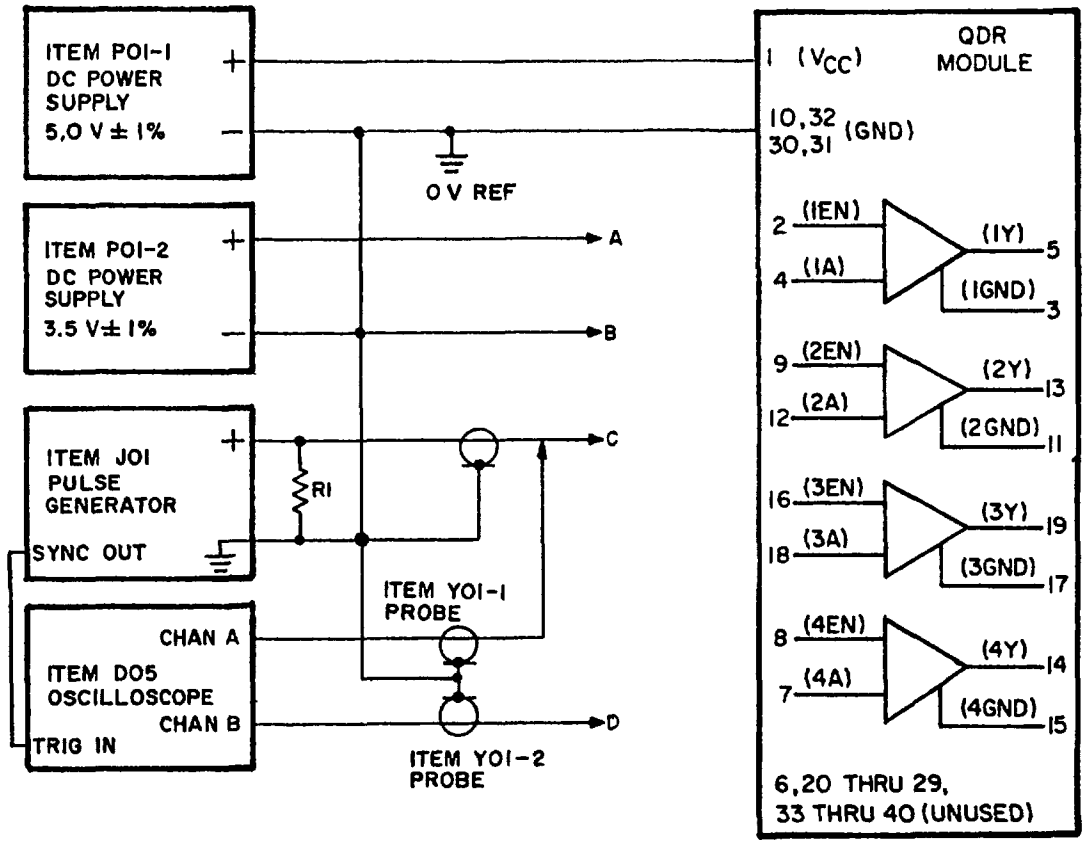
FIGURE 21. Enable recovery time test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1Y Output crosstalk, low level	2,9,16,8	4	12,18,7	5	0	400	mV	22,23
V2	2Y	2,9,16,8	12	4,18,7	13				
V3	3Y	2,9,16,8	18	4,12,7	19				
V4	4Y Output crosstalk, low level	2,9,16,8	7	4,12,18	14				
W1	1Y Output crosstalk, high level	2,9,16,8,4	Open	12,18,7	5				
W2	2Y	2,9,16,8,12	Open	4,18,7	13				
W3	3Y	2,9,16,8,18	Open	4,12,7	19				
W4	4Y Output crosstalk, high level	2,9,16,8,7	Open	4,12,18	14	0	400	mV	22,23

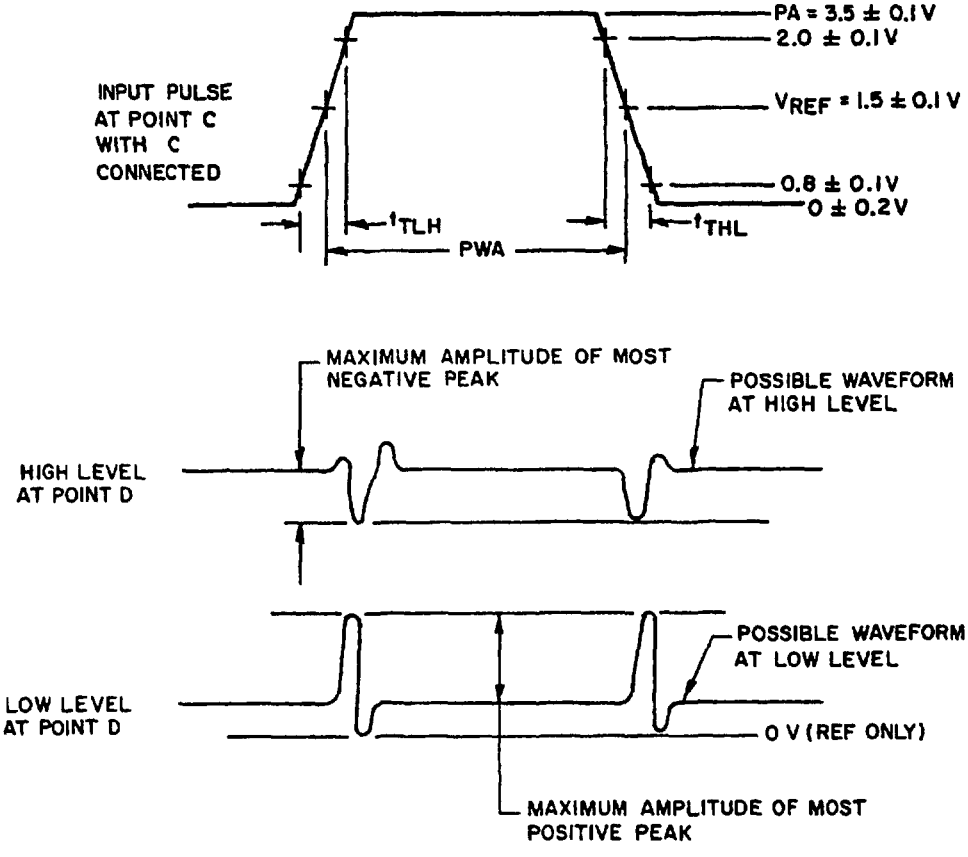
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 5.

FIGURE 22. Output crosstalk (low-level and high-level) test circuit for table XIX.

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NOTE: Input pulse parameters·
PWA = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%.

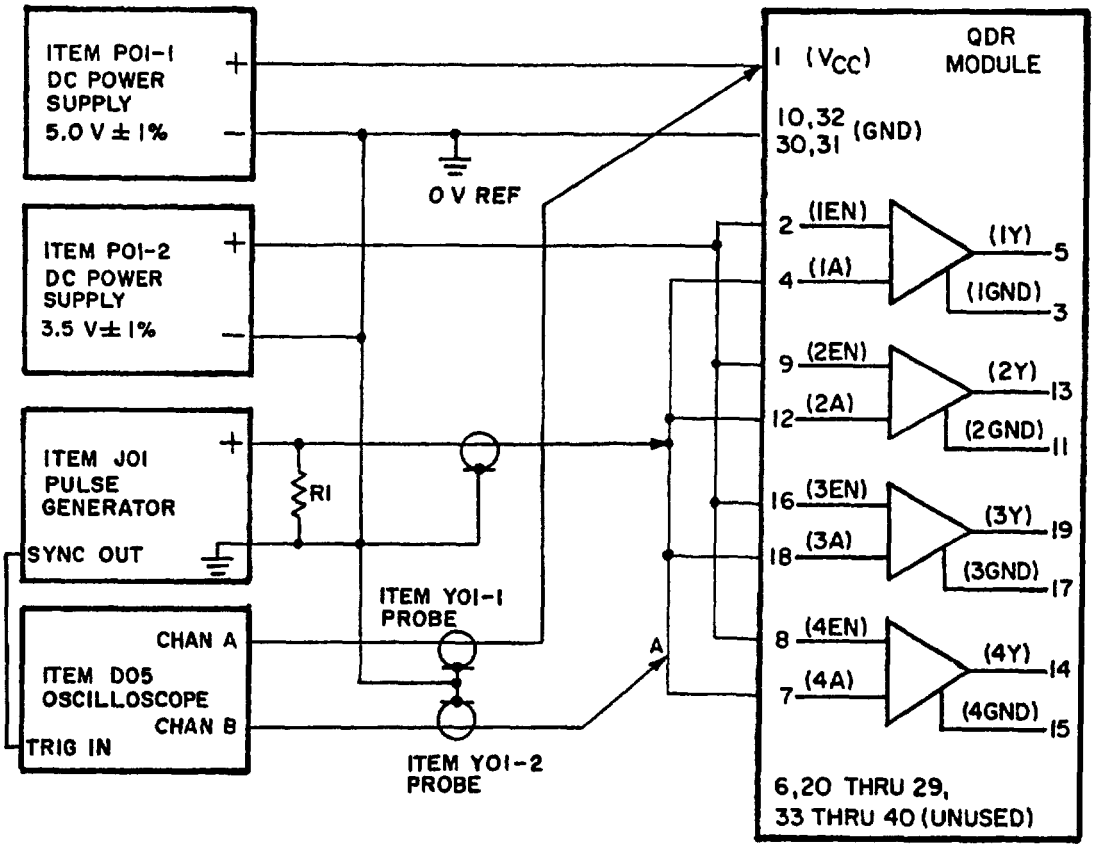
FIGURE 23. Output crosstalk test circuit waveforms for table XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	24,25

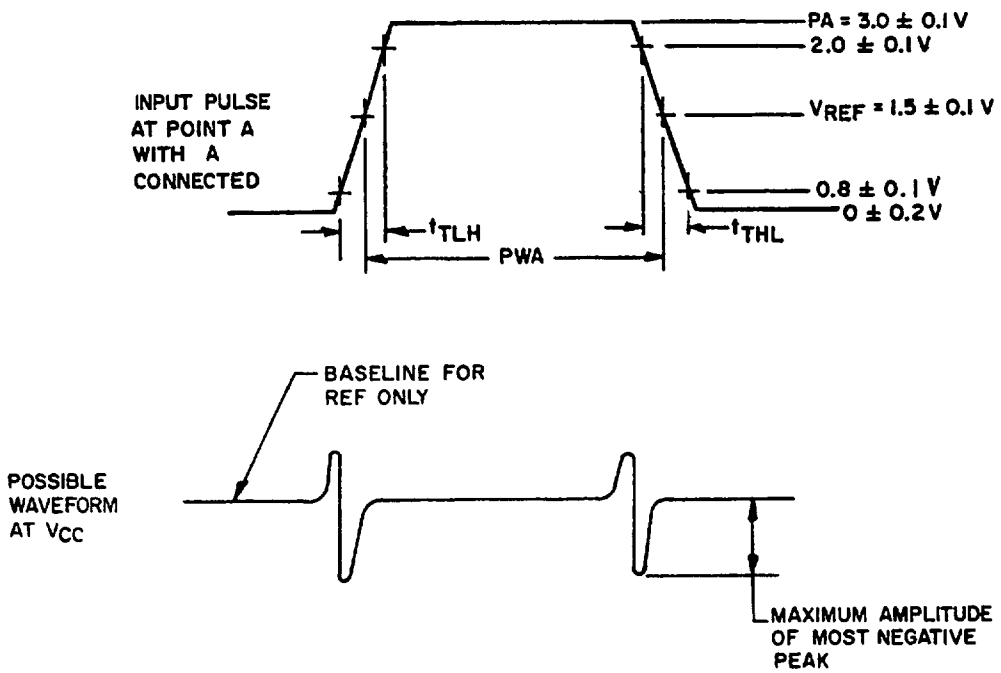
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 24. Power supply transient amplitude test circuit for table XX.

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NOTE: Input pulse parameters:
PWA = 500 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input).
PRR = 1 MHz ±10%.

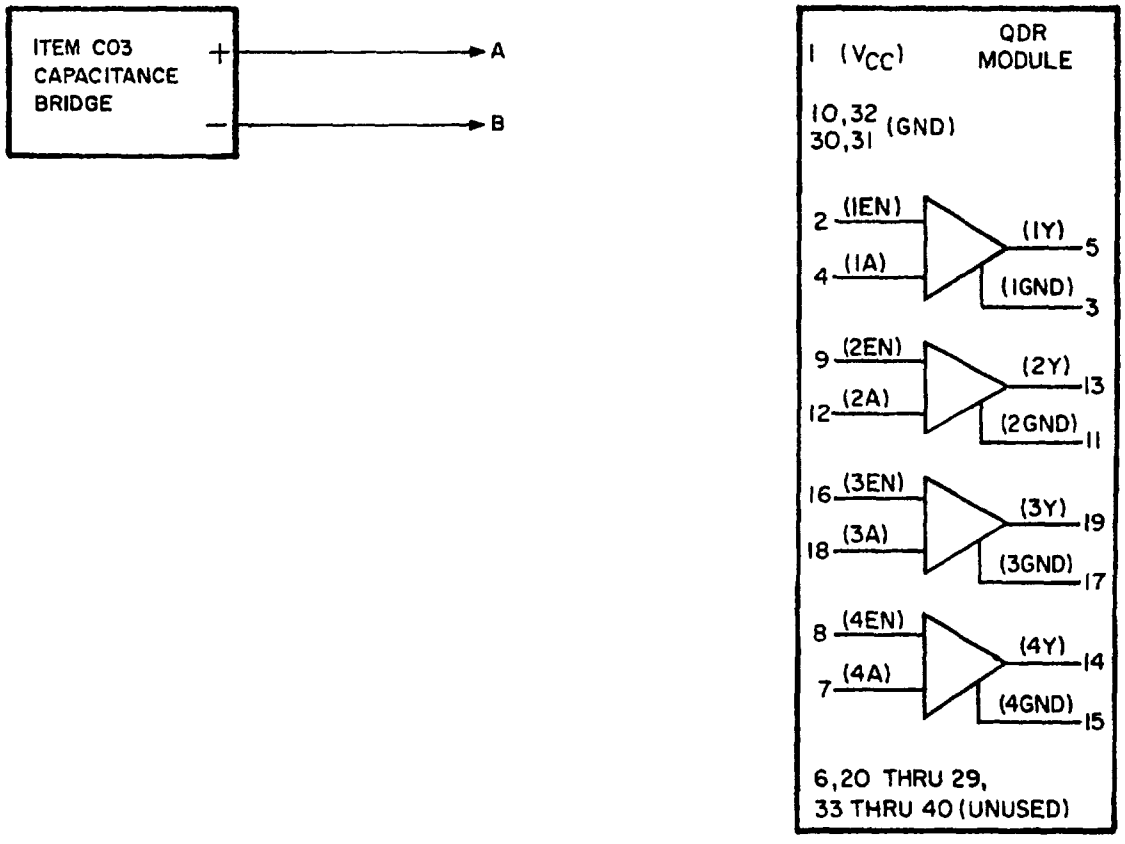
FIGURE 25. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	1	10,30,31,32	1.6	3.0	µF	26

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NOTE: Item CO3 setup:
V_{pk} = 0.3 V maximum.
Frequency = 120 Hz.

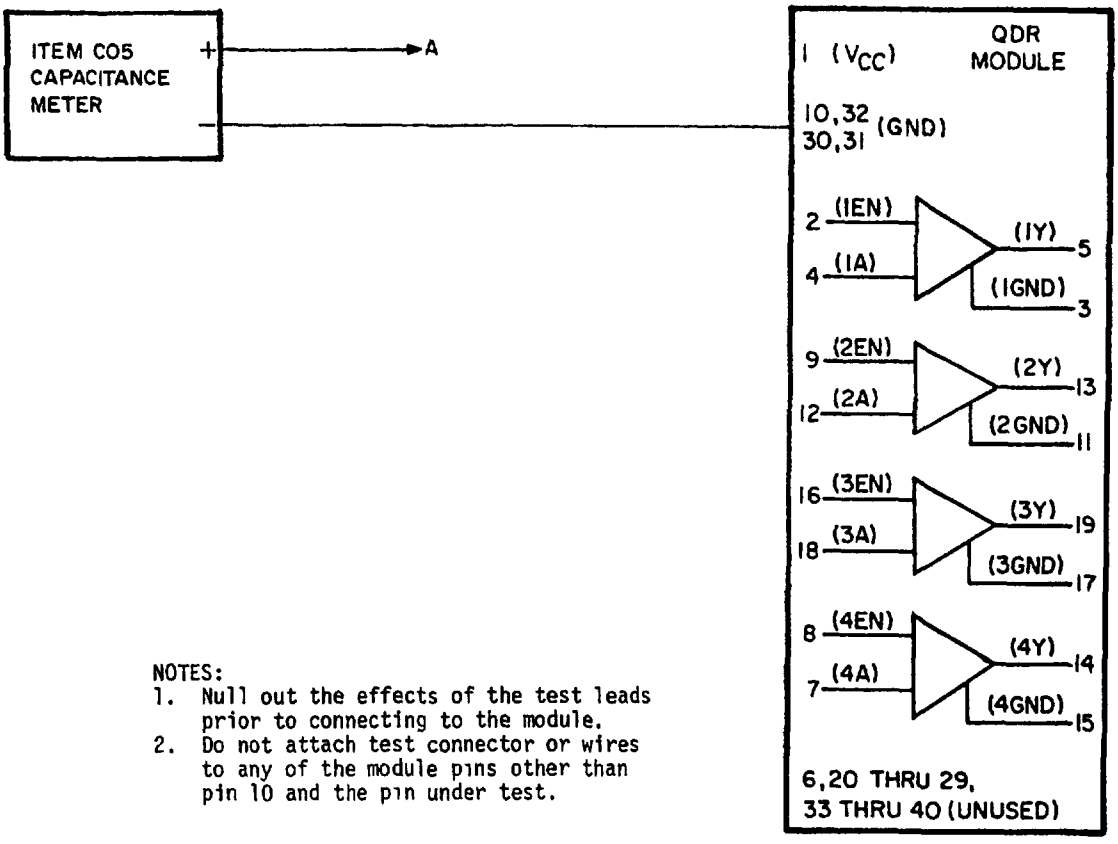
FIGURE 26. Power supply filter capacitance test circuit for table XXI.

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TABLE XXII. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Z1	1EN Input capacitance	2	3	15	pF	27
Z2	1A	4				
Z3	2EN	9				
Z4	2A	12				
Z5	3EN	16				
Z6	3A	18				
Z7	4EN	8				
Z8	4A Input capacitance	7	3	15	pF	27

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NOTES:
1. Null out the effects of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test.

FIGURE 27. Input capacitance test circuit for table XXII.

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